

Response to a Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Submitted by Lehigh University

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Introduction

The fundamental capabilities of Artificial Intelligence (AI) for expanding opportunities in U.S. industry, healthcare, education, and infrastructure are clear, but will not be met without leadership and investment from the federal government. We propose a series of fundamental principles that should guide the development and deployment of AI systems, and describe opportunities and challenges in several key areas. We believe that investments that integrate academic and private sector entities to push AI innovation forward will ensure U.S. leadership in AI. Innovation programs will spur investments in transportation, healthcare, infrastructure, national labs, and other areas, with a consistent focus on national security and on promoting US advances in AI growth. Ongoing innovation will require changes in policy and law, with ethical considerations given to the Constitutional rights of people and industries, and the goals of the United States.

1. Fundamental Principles Guiding AI Development and Deployment

A. Implementing Privacy Safeguards to Promote Trust and Adoption: Data protection is fundamental to national security in the AI era. Large-scale data is essential for AI development; however, if data—including citizens’ personal information and classified materials—is not adequately secured and regulated, malicious actors may exploit it. Effective protection of sensitive data reduces critical security risks, such as espionage, identity theft, and foreign influence operations. Experts caution that the commodification of personal data, without comprehensive privacy regulations, has generated “untenable national security threats” by allowing adversaries to collect and misuse U.S. citizens’ data.

Goal: To address these vulnerabilities, AI systems must incorporate robust privacy safeguards, including encryption, stringent access controls, and clear data-use policies, to prevent data breaches or cyber intrusions. Furthermore, ensuring privacy-respecting AI surveillance and data analytics enhances public trust—an essential factor for effective and sustained national security initiatives.

B. Promoting Transparency and Accountability for AI solutions: National security applications of AI—spanning military decision-support systems, intelligence analysis tools, and cybersecurity defenses—must be transparent, accountable, and consistent with ethical standards. Ethical AI deployment requires lawful and responsible usage, with human oversight for high-stakes decisions. For instance, the U.S. Department of Defense implemented AI ethical principles to align AI usage with American values and the laws of war. These principles emphasize that adopting AI for strategic advantage

does not compromise the United States' commitment to responsible and lawful conduct. Likewise, the Intelligence Community's ethics framework highlights transparency and accountability, pledging openness about AI deployments (within security constraints) and holding personnel accountable for AI-driven decisions (Intelligence.gov).

Goal: These requirements translate into a practical need to develop audit trails, interpretable outputs, and independent oversight to deter misuse. By adopting these ethical guidelines, defense and intelligence agencies can expand their capabilities while upholding legal norms and public trust, ensuring AI remains a principled force multiplier for national security.

C. Enhancing Legal and Regulatory Frameworks to Align with National Priorities:

Existing laws and regulations frequently lag behind the rapid advancement of AI technology, creating governance gaps that pose significant risks to national security. It is essential to establish a comprehensive legal framework for AI that aligns clearly with national security priorities and democratic principles. One major concern is the absence of comprehensive federal privacy legislation in the United States, a gap that experts highlight as having heightened security risks due to largely unregulated data practices.

Goal: Address this gap in order to establish uniform national standards for data protection in AI applications. Moreover, specialized governance for AI usage in national security contexts is required. This involves clearly defining prohibited or high-risk AI applications (such as autonomous lethal weapons without meaningful human oversight) and implementing robust risk management procedures for critical AI systems.

D. AI Data Center Efficiency: AI data centers are among the fastest-growing energy consumers globally, as AI workloads require significantly more computational power and cooling capacity than traditional data centers. Expansion of AI-driven infrastructures increases energy demand, strains electrical grids, and places new pressures on water resources due to cooling requirements. Unlike other infrastructures, such as residential communities, commercial buildings, and water distribution systems, data centers exhibit unique water-energy interdependence where cooling demands dictate water consumption, and energy use dictates dynamic AI processing loads. Traditional water-energy nexus models focus on distributed energy systems or municipal water distribution, but these systems are inadequate for AI data centers due to the rapid fluctuations in workload, cooling demand, and power intensity. Standard building-level or urban-scale water-energy optimization methods do not capture the unique real-time thermal fluctuations, spatial constraints of cooling systems, and the hyperscale energy density of data centers. The unique and demanding needs of AI centers require novel solutions.

Goal: Develop guidance for smart, efficient AI data center growth. With a significant increase in buildouts of AI data centers in the next decades, supporting industries and

local regions require guidance on design, operation, and infrastructure of these unique facilities, namely: AI data center infrastructure design; electricity grid upgrades, operations and dynamic adaptations; local energy generation and storage support; secure and efficient AI workloads co-designed with data center operations and maintenance, and underlying, novel cybersecurity concerns. This guidance and efficient AI data center solutions can be realized through research funding and workforce development programs that support partnerships among and between academic institutions, national labs, industries (e.g., AI data companies, utilities, ISOs, building construction and operations companies, OEMs, etc). A national program with regional centers of excellence that leverage such partnerships could address efficient AI data center solutions based on the region's energy portfolios, geographic and workforce features, economic development plans, etc. The Department of Energy, National Science Foundation (NSF), and/or Department of Commerce could be agencies to house such research and workforce development programs.

2: Opportunities and Challenges in Specific AI Application Areas

A: Transportation and Mobility

AI has already revolutionized transportation and mobility by enhancing efficiency, safety, and sustainability. AI-powered systems optimize traffic management, reduce congestion, and improve public transit through predictive analytics and real-time data processing. Autonomous vehicles, powered by AI, have the potential to minimize human error, significantly lowering accident rates and improving road safety. Additionally, AI-driven logistics and fleet management solutions streamline supply chains, reducing fuel consumption and emissions. Increased investments in AI for transportation are crucial to accelerating these advancements, fostering smart city development and addressing growing urban mobility challenges. Increased funding will drive innovation, improve infrastructure, and ensure AI-driven transportation solutions are accessible and sustainable for the future.

Goal: Establishing cross-sector partnerships for innovation. Establishing partnerships, collaborations, and joint projects can stimulate and innovate research in this area. An effective incentivization program to integrate academia and the private sector in transportation and mobility RD&D would focus on funding, collaboration, talent development, and regulatory support. Joint research grants, tax incentives, and venture capital funding could encourage universities and companies to co-develop AI-driven mobility solutions. Innovation hubs and technology transfer offices would streamline the pathway to commercialization, while industry-academia fellowships and specialized training programs would develop and ensure a skilled workforce. Regulatory sandboxes and public-private data-sharing frameworks would enable safe testing and deployment of smart transportation technologies. Performance-based funding and recognition

programs could further drive impactful solutions, accelerating advancements in autonomous vehicles, smart traffic management, and sustainable mobility systems.

B. Healthcare Innovation

Healthcare has already been transformed by AI. It enables faster diagnoses, personalized treatments, and more efficient medical workflows. AI-powered tools analyze vast amounts of medical data to detect diseases earlier, assist in medical imaging interpretation, and optimize drug discovery. Machine-learning algorithms enhance patient care by predicting health risks and personalizing treatment plans based on individual genetics and lifestyle factors. Additionally, AI-driven automation reduces administrative burdens, allowing healthcare professionals to focus less on paperwork and more on patient care. Increased investment in AI for healthcare can compound the payoff of the initial investment in research, accelerating medical breakthroughs, and in patient care, improving accessibility and reducing costs. Enhanced, next-gen medical AI can ultimately lead to better patient outcomes and a more efficient, data-driven healthcare system.

Goal: Medical advancements through commercialization. An effective incentivization program for integrating academia and the private sector in healthcare AI innovation would include joint research grants, tax incentives, and startup funding to drive collaboration. Establishing innovation hubs, technology transfer offices, and industry-academia fellowships would accelerate knowledge sharing and ramp up commercialization efforts. Specialized AI-in-healthcare training programs and workforce exchanges would ensure talent development. Regulatory support, including fast-track approvals and secure data-sharing frameworks, would facilitate innovation while ensuring compliance. Performance-based incentives and recognition programs would further drive impactful outcomes. This holistic approach would accelerate AI-driven healthcare advancements by leveraging the strengths of both academic and private sector entities.

C. Modernizing Control and Operations of Energy Systems

The last two decades of upgrades to national energy systems, particularly electricity grids, have enabled more comprehensive, high-resolution, and often real-time data collection capabilities in all aspects of energy conversion, transmission, distribution, and consumption. These capabilities can be leveraged through AI-powered control and monitoring systems to improve the efficiency, security, and resilience of energy systems and to enable dynamic interactions with energy end-users. Because of the criticality of the services provided by energy systems, modifying their business-as-usual operations will require significant financial and regulatory support in order to research, develop, test, implement, and scale these AI-enabled solutions while ensuring their performance benefits and security systems.

Goal: Deeper penetration of AI in energy systems. Effective mechanisms to enable deeper penetration of AI solutions in energy systems will involve partnerships between research entities (e.g., academic institutions, national labs), industry (e.g., utilities, ISOs, OEMs, data companies), and consortia and standards boards (e.g., EPRI, GTI Energy, IEEE) to design, pilot, test, and scale AI-driven solutions. These partnerships will require joint research, workforce development, and startup funding (e.g., through programs in Department of Energy, ARPA-E, National Science Foundation, and National Institute of Standards and Technology) to demonstrate viability of AI-powered energy systems. Additionally, regulatory support is needed to incentivize testing and integration of these solutions while safeguarding reliable delivery of critical energy services, privacy of consumer data, and economic decision-making.

D. Cybersecurity in AI Models and Infrastructure

A comprehensive action plan focusing on fundamental, interdisciplinary, and translational research. AI systems have become integral to numerous sectors, enhancing efficiency and decision-making processes. However, their increasing integration presents significant cybersecurity challenges, particularly concerning the integrity of AI models and the resilience of critical infrastructures like power grids. The consequences of cyberattacks exploiting AI vulnerabilities are profound; adversaries could manipulate AI systems to disseminate misinformation, conduct sophisticated phishing schemes, or disrupt essential services, leading to economic instability and societal harm. For instance, AI-generated deepfakes have been used to create disinformation and fraud, while large language models have facilitated the creation of fake reviews to manipulate e-commerce ratings. Moreover, the potential for AI to be harnessed in cyber warfare is a growing concern, as evidenced by reports of nation-states developing AI capabilities to enhance their cyber-attack strategies. Addressing these concerns necessitates a comprehensive action plan focusing on fundamental, interdisciplinary, and translational research. Cybersecurity is crucial to national security as well as to trust, adoption, and monetization of these technologies.

Fundamental Research on Security Issues in AI Models

Addressing security vulnerabilities in AI models during both training and inference phases is crucial for developing resilient AI systems. During training, data poisoning attacks involve contaminating the training dataset with malicious data, effectively reprogramming algorithms with potentially harmful intent. This is particularly concerning for models trained on user-generated data, as adversaries can exploit fake accounts to introduce misleading information, thereby compromising the model's integrity. For instance, disinformation campaigns on social media attempt to bias recommendation and moderation algorithms to push certain content over others. During inference, adversarial attacks pose significant threats, especially to large-scale models like Large Language Models (LLMs). These attacks involve crafting inputs designed to deceive AI

systems into making incorrect predictions or classifications. For example, prompt injection attacks can manipulate LLMs into generating harmful content or revealing sensitive information. The complexity and vastness of training data in state-of-the-art models make distinguishing between legitimate instructions and adversarial inputs particularly challenging, thereby increasing their susceptibility to manipulation. To mitigate these threats, research is focusing on enhancing model robustness through adversarial training, developing sophisticated input validation techniques, and implementing continuous monitoring systems to detect and respond to unusual model behaviors in real-time. These efforts are essential to safeguard AI systems against both data poisoning and adversarial attacks, ensuring their reliability and security in various applications.

Interdisciplinary Research between AI Security and Power Grid Infrastructure

Interdisciplinary research bridging AI security and power grid infrastructure is vital, given the escalating energy demands from AI data centers. AI workloads require significantly more computational power and cooling capacity than traditional data centers, making them one of the fastest-growing energy consumers globally. According to the U.S. Department of Energy, U.S. data centers are projected to consume between 325 and 580 terawatt-hours (TWh) per year by 2028, accounting for 6.7% to 12% of the nation's total power consumption. Research has demonstrated that AI models can consume up to 67% more power when input data is specifically crafted to maximize dynamic switching power in the underlying VLSI circuits, even when all other factors—such as the AI model, data size, and hardware—remain unchanged. If an adversary, disguised as a benign user, exploits this phenomenon at scale, they could induce sudden power surges in AI data centers, potentially leading to service disruptions or physical damage. Beyond impacting data centers, such power surges place immense strain on the power grid. Given that cybersecurity measures in power dispatch systems are often outdated, adversaries may coordinate cyberattacks on the power grid in tandem with AI-targeted attacks, reducing the grid's ability to respond effectively. Such an event could trigger cascading failures, including widespread blackouts, with devastating consequences for national security and economic stability. Collaborative research efforts are crucial to developing coordinated defenses that protect both AI systems and power grid infrastructure from these emerging cyber threats.

Goal: Translational Research on Cybersecurity for AI. To bolster translational research in AI cybersecurity, federal agencies like the NSF should implement strategic initiatives that bridge foundational research with practical applications. Establishing dedicated funding programs that encourage interdisciplinary collaborations among academia, industry, and government entities is essential. Additionally, agencies should facilitate public-private partnerships to ensure that research outcomes are effectively integrated into real-world systems, enhancing the resilience of AI technologies across various sectors. Investing in educational initiatives to cultivate a workforce proficient in

both AI and cybersecurity is also crucial, ensuring that emerging challenges are met with well-equipped professionals. By adopting these measures, federal agencies can effectively translate foundational research into robust cybersecurity solutions for AI systems. To further advance translational research in AI cybersecurity, the National Science Foundation (NSF) could issue a Dear Colleague Letter (DCL) to highlight the importance of this area and encourage researchers to engage in related projects. By publishing a DCL on AI cybersecurity, the NSF would signal a strategic emphasis on this critical field, fostering increased awareness and stimulating innovative research endeavors. This initiative would complement existing funding programs and interdisciplinary collaborations, ensuring that cybersecurity considerations are seamlessly integrated into the development of AI technologies.

E. Improving AI Adoption in Critical Decision Making through Explainable and Accountable Solutions

Despite vast amounts of data available, real-time data-driven decision-making during extreme events such as natural disasters or large-scale cyber-attacks continues to be challenging. Emergency responders often rely on experience or on pre-engineered solutions with limited usage of real-time data. Although faster emergency responses powered by AI solutions have the potential to lower the financial and human costs of disasters, the adoption of real-time data-driven solutions has been minimal. There are important technical and social reasons for this lag. From a technical perspective, existing predictive AI solutions work exceedingly well under normal conditions, but are unreliable in the extreme conditions that are unrepresented in AI model training data. Furthermore, an extreme event causes rapid changes in behavior of the underlying system and its human participants, which requires fast adaptability of solutions. From a social perspective, in critical events, emergency response actions are taken by human operators who bear professional liability for their decisions. AI solutions that are produced by complex and opaque computational models, without support of logical reasoning, are highly unlikely to be adopted in mission critical situations. Addressing these challenges requires a concerted effort that brings together academic, industrial, and public policy efforts.

Goals: Predictive AI, Risk-Adaptive Real-Time Computation, and Explainable AI as explained below.

Basic Research in Explainable AI:

- *Predictive AI for unseen and unpredictable conditions:* A major challenge in ML, is the development of models that provide reliable outcomes when test data patterns are not seen during the training phase; this is referred to as out-of-distribution generalization. Despite various approaches designed for ML in research areas such as vision and text, addressing the challenge in dynamical and cyber-physical systems remains open. For infrastructural systems, we require basic research in domain-informed machine learning

and innovative computational architectures that adhere to underlying physical laws to improve prediction performance under conditions significantly different from those encountered in training data. Outcomes from such domain-informed computation are inherently explainable to operators using them.

- *Risk-Adaptive Real-Time Computation under rapidly evolving conditions.* Extreme events present scenarios with rapidly evolving risks of damage and disruption. Computational algorithms for decision-making in systems are typically designed based on static or predictable system states, and are not trained to include information such as impending risks of damage from an evolving disaster or a cyberattack. Basic research in catastrophe modeling offers the opportunity to assess in near-real-time the probability of loss of functionality of system components and processes. This, alongside the development of new risk-aware computational tools can greatly enhance adoption of data-driven decision-making in rapidly evolving critical events.

- *Explanations for computational outcomes:* From a cognitive psychological perspective, very little is known about what kind of explanations for data-driven models can gain trust of actual users of the methods and outcomes. Explainable AI methodologies that have been developed, particularly for machine learning solutions, are focused on “simplifying” the computational process or outcomes rather than considering a holistic view of the user in the context of the application. In place of a computation-only approach, what is required is a holistic framework to derive explanations that are computationally accurate, contextually interpretable, and guided by a use-inspired framework of human perception and decision-making. This requires basic research to bring together concepts and tools from knowledge representation, cognitive psychology, data science, optimization, and machine learning, to build explainable representations of domain knowledge, derive rules to contextualize the outcomes of data-driven decisions, and use cognitive psychological studies to formally understand the perception and use of explanations during extreme events to help formulate explanations that will be adopted and used.

Translational Research for Deployable AI Solutions

The biggest gap between basic research in AI and deployable solutions for engineering systems is the availability of real-data for developing, validating, and refining computational methodologies. As a result, companies seeking AI solutions that can afford to invest in in-house basic research do so, while many others have no choice but to use off-the-shelf technologies that are highly sub-optimal. To promote better translational research for AI solutions, there is a need to lower the “data barrier” between basic research and deployable AI. We recommend the following actions:

- Promoting industry-academic partnerships targeting mission-critical AI through creation of research funding programs
- Creation of Secure Data Repositories for companies to share data with academic partners ensuring that the needed privacy protections are in place.

- Investment in joint industry-academic workforce training programs at the postgraduate level where participants can gain foundational research expertise in AI while also gaining the experience in translating research ideas in practice with real-world data and scenarios.

Policy Development

Even as public policy around AI is evolving, situations involving critical decision-making in which liability is an important factor require strong safeguards that can be better facilitated through targeted efforts to improve accountability of AI solutions. Specifically:

- Basic research in determining the “failure modes” for AI solutions and quantifying the costs associated with these failure modes.
- Development of standards for performance of AI solutions, not merely in computational metrics such as correctness, speed, efficiency, but also on failure costs, ethical violations, and usability.
- Defining policy for better communication around AI solutions: Clear communication about AI solutions requires tailoring explanations to different audiences, defining key concepts precisely, and avoiding misleading claims. Emphasizing transparency, clear assessment of risk, and accountability for failure modes, helps build trust, limits liability, and ensures responsible AI adoption.

Conclusion: In order to continue and expand U.S. primacy in AI, we want to focus on opportunities to bring AI technologies and capabilities to the next levels for industry and academia, as well as the nation overall. Exploring and advancing AI opportunities in transportation, healthcare research and administration, energy systems, cybersecurity, and critical decision making will require multidisciplinary approaches as well as partnerships across industries, academia, national labs, and even international partners. Incentivizing partnerships in all these areas will promote AI innovation and growth, as well as American leadership, in AI in all areas of life and business.